Fiscal Year:	FY 2008	**Task Last Updated:**	FY 07/16/2008
PI Name:	Durian, Douglas J. Ph.D.		
Project Title:	The Melting of Aqueous Foams		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19104-6396	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	1995 Microgravity Fluid Physics NRA-94-OLMSA-05: Research and Flight Experiment Opportunities
Start Date:	08/05/2004	**End Date:**	04/04/2010
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: Funding extended to 4/4/2010 through a new grant in 2007 (6/2008) NOTE: Funding reinstated thru 9/30/2008, per F. Chiaramonte (8/2006). Suppl funding thru Oct 2006 (6-5-06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX07AK60G ; NNC04GB61G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The unusual elastic character of foams will be quantified macroscopically by measurement of the shear stress as a function of static shear strain, shear strain rate, and time following a step strain; such data will be analyzed in terms of a yield stress, a static shear modulus, and dynamical time scales. Microscopic information about bubble packing and rearrangement dynamics, from which these macroscopic non-Newtonian properties presumably arise, will be obtained non-invasively by novel multiple-light scattering diagnostics such as diffusing-wave spectroscopy (DWS) and speckle-visibility spectroscopy (SVS). Bubble size and homogeneity will be studied by diffusely transmitted light (DTS). Liquid fraction will be measured by electrical conductivity. Quantitative trends with materials parameters, such as average bubble size, and liquid content, will be sought in order to elucidate the fundamental connection between the microscopic structure and dynamics and the macroscopic rheology. There are three all-encompassing objectives motivating the FOAM project. The first concerns multiphase flows and represents the “mechanics” aspect alluded to in the F.O.A.M. acronym: To understand how foams and bubbly liquids deform and flow in response to applied forces. The second objective concerns multiphase separations: To understand how foams and bubbly liquids evolve with time via a combination of liquid flow between bubbles, of gas diffusion across the film separating neighboring bubbles, and of direct coalescence via film rupture. For both flows and separations, some level of understanding has been established for very dry foams, and also for liquids with relatively few bubbles; however, we are more ignorant of the crucial intermediate regime of very wet foams and very bubbly liquids. This regime defies earth-bound experiments. The third, overall, objective is to transfer this new understanding of multiphase flows and separations into ASL technology for NASA's strategic exploration mission. The utility and fascination of foams are derived largely from the surprising fact that they have a solid-like elastic character in spite of being mostly gas with a few percent volume fraction of liquid, but can nevertheless flow under shear. The physical origin of such unusual rheology in terms of microscopic structure and dynamics is poorly understood and remains a subject of basic scientific interest to physicists, chemists, and chemical engineers. The proposed research promises important new insight into these issues, and could also have significant consequences for our understanding of flow in other dense randomly-packed systems such as emulsions, colloidal suspensions, slurries, bubbly liquids, and granular materials. Furthermore, all foam applications are empirically based and the proposed research may generate valuable fundamental guidance for the development of materials with more desirable rheology and better stability. The value of the FOAM project in terms of NASA’s exploration mission lies with Advanced Life Support. Water recovery and recycling are absolutely central to ASL, and require fine control of multiphase flows and separations. The very heart of the FOAM project is to bring new technology and understanding to these problems
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Foams are important for a clean environment. Firstly, their presence in natural waters signals the presence of surface-active impurities and a need for clean-up. Secondly, they are useful themselves in the cleanup process since many impurities can be removed by injecting gas and skimming off the resulting foam. Relatedly, study of foams brings specific insight into multiphase flow and separation phenomena of direct importance for water recovery and recycling operations.
Task Progress:	During this period my group included Alex Gittings, Adam Roth; Matthew Berck, and Dr. Klebert Feitosa, and Dr. Chris Jones. Alex is a graduate student, who will graduate soon; Adam Roth is a beginning graduate student who just joined to replace Alex; Klebert is a postdoc, whose term expired September 2007; Chris is a postdoc who began in November 2007; Matthew is an undergraduate physics major. During the past year we have made significant progress on several different aspects of the overall FOAM project: (1) I attended a meeting at ESA headquarters in June, where we redefined the scope of the FOAM mission in order to align with cost and engineering limitations. At my urging the mission now will focus on the physics of coarsening, and the dependence of the coarsening rate on the liquid fraction. Rheology aspects will be postponed for future flights. In consequence I helped formulate a new ESR (Experimental Requirements Document). The experiment features optical probes, like DTS, DWS, and now SVS as crucial diagnostics of the structure and dynamics. (2) I gave an invited presentation on bubble rearrangements and SVS at the annual colloid and interface science meeting of the American Chemical Society. And I gave a presentation on both SVS and higher-order intensity correlation methods, developed as part of the foam project, for characterizing unsteady dynamics at a Brookhaven National Laboratories conference on x-ray photon correlation spectroscopy. (3) I gave an invited presentation on unsteady motion is coarsening, sheared, and oscillating foams at a conference on jamming in soft matter at the Lorentz Institute in Leiden, Netherlands. (4) Steady-State Foams. As described in previous reports and publications, we produce foams in steady state by continuous bubbling of gas into a pool of surfactant solution at the bottom of a tall column. Our interest has been in time evolution via both drainage and coarsening, here combined in a novel previously unstudied way that also happens to be relevant to industrial processes like froth flotation for chemical and mechanical separations. This aspect of study, which was led by Klebert, is coming to a close, with a final publication written and just accepted for publication. Besides this new work on dissemination of results, we have begun a new project with Matthew to investigate the mechanics of the foam. In particular we now drop a plastic bead into the steady foam, with size comparable to the bubbles, and image its behavior vs time. The key observation is that the bead height fluctuations around a steady-state average value. This height, and the know density difference and gas flux speed, give the mobility of the particle – ie a measure of the foam mechanics. Furthermore, fluctuations around the average are related to bubble-scale dynamical rearrangements, and potentially may yield frequency-dependent rheological properties of the foam. Ultimately this new knowledge will be key for helping define the next set of flight experiments on rheology vs liquid fraction. (5) Speckle-Visibility Spectroscopy. As described in previous reports, this focuses on microscopic bubble-scale motion that fundamentally underlies the unusual flow properties of foams. The method is essentially a time-resolved version of DWS, capable of following dynamics that change or evolve with time. By contrast, DWS relies on temporal averaging. Whereas DWS can reveal bubble rearrangement rates, SVS can also reveal bubble speeds and the duration of motion during individual rearrangement events. During this cycle, Alex has finished analyzing a massive data set on an

	evolving foam, with the final step being to understand the baseline of the signal. Alex has now prepared a complete set of figures and written a rough draft. I have rewritten the paper, and now submitted it for publication. (6) Tomography. Klebert and Adam together have begun a new project where we hope to obtain high-quality three-dimensional images of dry foams. Such data will form the basis for understanding structure at high liquid fractions, to be obtained in microgravity. Surprisingly, the best such images still date from 1945 on monodisperse foams. Our interest, of course, is to obtain even better data for polydisperse foam which have evolved by coarsening into a stationary state where the size distribution has a fixed shape and is independent of initial conditions. Toward this end, we have purchased a high-quality lens, a rotation stage, and lighting. We have assembled, aligned, and optimized the setup. We have created analysis programs. However these are slow and need to be improved; Adam has begun this work. This is now a time of investment, as Adam takes classes and begins to devote more and more of his time to research. (7) New directions in the laboratory. The new postdoc, Chris Jones, will assist Adam Roth with the tomography project but will also explore new avenues for ground-based support of the flight project. To begin he will explore packings of only a few bubbles made from a liquid crystal so that the defect structure can be viewed using polarization methods. This will take advantage of Chris' expertise from his PhD thesis, and will allow us to make contact with an unpublished report that foam made from liquid crystals exhibit anomalously slow coarsening. In addition, Chris has also begun a novel rheology experiment for very dry foams. The idea is to use PIV methods to capture the velocity flow profile for foam in a vertical column, driven by bubble production from below. Viscosity vs strainrate can be deduced from measurements of velocity vs radial position and the known pressure drop across the sample, by simple force balance.
Bibliography Type:	Description: (Last Updated: 06/12/2025)
Abstracts for Journals and Proceedings	Durian DJ. "Bubble rearrangements statistics probed by speckle visibility spectroscopy." Presented at Annual Colloid and Interface Society Meeting of the American Chemical Society, Wilmington DE, June 2007. Annual Colloid and Interface Society Meeting of the American Chemical Society, Wilmington DE, June 2007. , Jun-2007
Abstracts for Journals and Proceedings	Durian DJ. "New dynamic light scattering probes for heterogeneous / unsteady dynamics." Presented at NSLS-II Workshop on X-ray Photon Correlation Spectroscopy, Brookhaven NY, January 2008. NSLS-II Workshop on X-ray Photon Correlation Spectroscopy, Brookhaven NY, January 2008. , Jan-2008
Abstracts for Journals and Proceedings	Durian DJ. "Review of unsteady motion in flowing, coarsening, and oscillating foams." Presented at Lorentz Center Workshop on Jamming in Soft Matter under Driving – Colloidal Suspensions, Granular Media, Foams, and Complex Plasmas, Leiden Netherlands, February 2008. Lorentz Center Workshop on Jamming in Soft Matter under Driving – Colloidal Suspensions, Granular Media, Foams, and Complex Plasmas, Leiden Netherlands, February 2008. , Feb-2008
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